University of Utah Fall 2025

MATH 3220-002 PSet 3 Specification

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1 Background

This problem set focuses on metric spaces. The problems are designed to guide you through three key areas:

1. Developing intuition for convergence in normed spaces (Problems 1–2).
2. Introducing the idea of metric spaces as geometric objects (Problems 3–4).
3. Connecting the geometry of metric spaces back to the structure of normed spaces (Problem 5).

For metric geometry, first we establish the notion of the length of a path in a metric space (which generalizes the classical notion of length of a path as the integral of the speed of the path), and then, using the notion of lengths of paths, we intrinsify a metric by infimizing over all paths.

For instance, take two points on the unit circle in the plane. One could declare their distance to be the standard Euclidean distance, corresponding to the length of the line segment connecting them. Intrinsifying this metric then corresponds to instead declaring their distance to be the length of the shorter segment of the circle that connects the two points.

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The hyperlinks too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.

2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.

3. **Construct a proof:** Provide a formal argument to support your conclusion.

4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical error detection and correction skills.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Consider the following sequences $x_\bullet$ of tuples of real numbers:

$$\mathbf{I}\colon\ n \mapsto \left(\frac{n^3+n-1}{3n^2+2},\ \frac{n-1}{n+1},\ \frac{\sin(n)}{2^n} \right)$$

II: $n \mapsto \left(1+(-1)^n \ , \ \dfrac{1}{n} \ , \ 1+\dfrac{1}{n} \ , \ \dfrac{\cos(n)}{3^n} \right)$

III: $n \mapsto \left(\log(n+1) - \log n \ , \ \sin\left(\dfrac{1}{n}\right)\right)$

For each sequence $x_\bullet$, perform the following tasks:

(a) Determine whether or not $x_\bullet$ is convergent.

(b) If $x_\bullet$ is convergent, compute its limit.

(c) Determine whether or not $x_\bullet$ is bounded.

(d) Determine a convergent subsequence of $x_\bullet$ and compute the limit of this subsequence.

2. Let V be a normed space and $v_\bullet$ be a sequence in V converging to 0. Then for any normed space W and any bounded sequence of bounded linear operators $B_\bullet$ from V to W,

$$\lim_{n\to\infty} B_n(v_n) = 0.$$

3. **This problem is a prequel to the next two problems.** Let (X, d) be a metric space. A **continuous parameterized path** (**path** for short) in X is by definition a continuous function $\gamma : I \to X$, where $I \subseteq \mathbb{R}$ is a closed and bounded interval. In this case we also say that γ is a path **from** $\gamma(\min(I))$ **to** $\gamma(\max(I))$. For a path γ, define the **length** of γ by

$$\operatorname{len}(\gamma) = \sup\left\{ \sum_{i=1}^{n} d(\gamma(t_{i-1}), \gamma(t_i)) \ \middle|\ \substack{n\in\mathbb{Z}_{\geq 1},\ \min(I)=t_0,\ t_n=\max(I) \\ t_0 \lneq t_1 \lneq \cdots \lneq t_{n-1} \lneq t_n} \right\}$$
$$\in [0, \infty].$$

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(a) Let $\gamma : [\lambda, \rho] \to X$ be a path and $\mu \in [\lambda, \rho]$. Then

$$\mathrm{len}(\gamma) = \mathrm{len}(\gamma|_{[\lambda,\mu]}) + \mathrm{len}(\gamma|_{[\mu,\rho]}).$$

Here $\gamma|_{[\lambda,\mu]}$ is the restriction of γ to the subinterval $[\lambda, \mu]$, and similarly $\gamma|_{[\mu,\rho]}$ is the restriction of γ to $[\mu, \rho]$.

(b) Let $\gamma : [\lambda, \rho] \to X$ be a path in X. Then the function $\Phi : [\lambda, \rho] \to [0, \mathrm{len}(\gamma)]$ defined by

$$\Phi : t \mapsto \mathrm{len}\left(\gamma|_{[\lambda,t]}\right)$$

is continuous.

(c) Let $\gamma : I \to X$ be a path in X. Let further $J \subseteq \mathbb{R}$ be a closed and bounded interval and $\Phi : J \to I$ be a monotone continuous function. Then

$$\mathrm{len}(\gamma) = \mathrm{len}(\gamma \circ \Phi).$$

4. **This problem is a sequel to the previous problem and a prequel to the next problem.** Let (X, d) be a metric space. A path $\gamma : I \to X$ is called **rectifiable** (or **of bounded variation**) if its length is finite.

Define a new function $d_{\mathrm{int}} : X \times X \to \mathbb{R}_{\geq 0}$ as follows: for any $x_1, x_2 \in X$, $d_{\mathrm{int}}(x_1, x_2)$ is the infimum of all the lengths of rectifiable paths from x_1 to x_2. d_{int} is called the **intrinsic metric induced by** d, and the metric transform $d \mapsto d_{\mathrm{int}}$ is called **intrinsification**.

(a) d_{int} is indeed a metric on X.

(b) $d = d_{\mathrm{int}}$ iff for any $x_1, x_2 \in X$ and for any $\varepsilon \in \mathbb{R}_{>0}$, there is a rectifiable path γ from x_1 to x_2 such that

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$$\mathrm{len}(\gamma) < d(x_1, x_2) + \varepsilon.$$

(c) $(d_{\mathrm{int}})_{\mathrm{int}} = d_{\mathrm{int}}$, that is, intrinsification is idempotent.

(d) A sequence in X converges relative to d iff it converges relative to d_{int}.

5. **The previous two problems are prequels to this problem.** Let X be a normed space and d be the metric induced by the norm. Define for any $x_1, x_2 \in X$, define $\lambda_{x_1,x_2} : [0, 1] \to X$ by

$$\lambda_{x_1,x_2} : t \mapsto (1-t)x_1 + tx_2.$$

(a) For any $x_1, x_2 \in X$, λ_{x_1,x_2} is a rectifiable path and $d(x_1, x_2) = \mathrm{len}(\lambda_{x_1,x_2})$.

(b) For any $x_1, x_2 \in X$, if $\gamma : I \to X$ is a rectifiable path from x_1 to x_2 with $d_{\mathrm{int}}(x_1, x_2) = \mathrm{len}(\gamma)$, then $\gamma = \lambda_{x_1,x_2}$.

(c) $d_{\mathrm{int}} = d$.

(d) Let $A \subseteq X$ and d_A be the subspace metric on A. Then $(d_A)_{\mathrm{int}} = d_A$ iff A is convex, that is,

$$\forall x_1, x_2 \in X, \forall t \in [0, 1] : \lambda_{x_1,x_2}(t) \in A.$$

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a generative AI tool or a computer algebra system for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or
- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., ChatGPT) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the Wayback Machine (see their "Save Page Now" documentation) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as prompt engineering, and it is your responsibility you design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

  Here are some guides regarding prompt engineering:

  – https://platform.openai.com/docs/guides/prompt-engineering
  – https://developers.google.com/machine-learning/resources/prompt-eng
  – https://www.ibm.com/topics/prompt-engineering
  – https://aws.amazon.com/what-is/prompt-engineering/

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

  https://forms.gle/Ch4N3PqneLMtEQ8D8.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

  https://www.gradescope.com/courses/1073533/assignments/6451786,

  see the Gradescope documentation for instructions.

  You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on September 19, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.